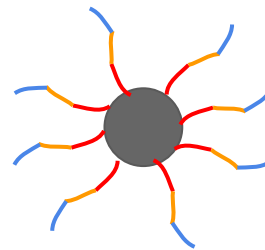
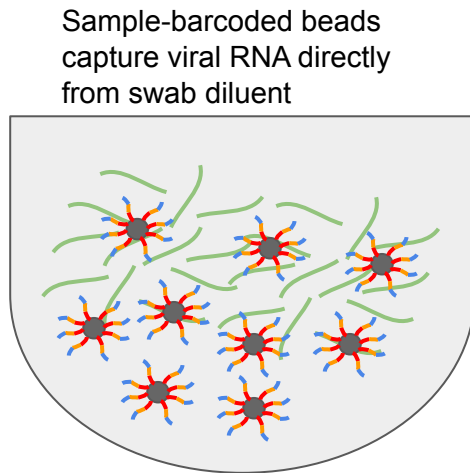


Direct-from-swab Illumina library prep

Credits: inspired by twitter discussion of Hossain *et al* 2020 protocol, using ARTIC network primalscheme, Bruce *et al* and Beltran-Pavez *et al* qPCR without extraction

This direct-from-swab Illumina library prep protocol avoids the need for per-sample RNA extraction and reverse transcription, thereby eliminating supply chain bottlenecks and significantly reducing per-sample costs. Up to $384^3 = \sim 56$ million samples can be processed in a standard 384-well plate format. Capture probes tile the entire ~ 30 kbp SARS-CoV-2 genome.



Barcoded magnetic bead with SARS-CoV-2 capture probes.
Red: Illumina adapter stub
Orange: sample barcode
Blue: nCov target



Viral RNA fragments

Workflow overview

